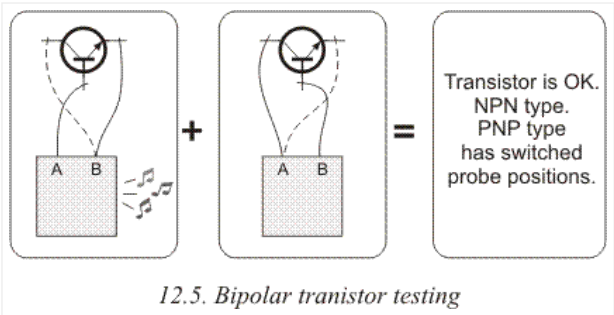
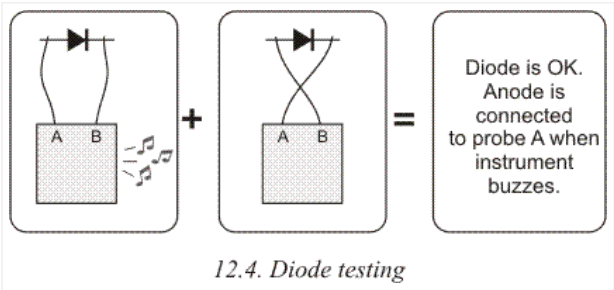


12.1 SEMICONDUCTORS CHECK

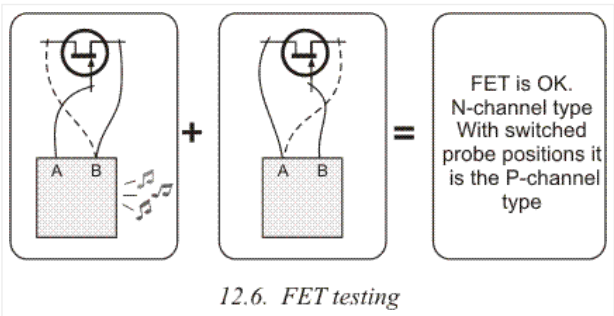
MIKROELEKTRONIKA

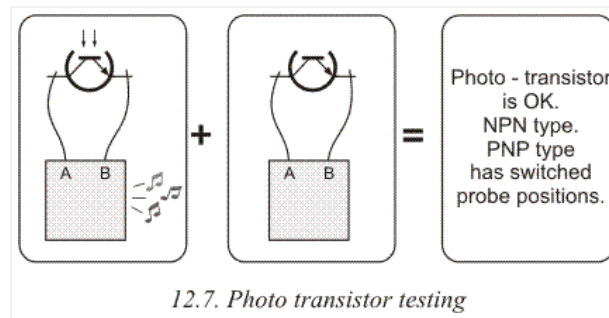
To test diodes using this circuit, we fall back to the diode theory of operation: when anode is positive comparing to the cathode (red probe on anode, black on cathode), whole diode acts as a low value resistor, which means that speaker sound is higher than usual. On the other hand, in the opposite direction, sound is lower because in that direction diode acts as a high value resistor. Testing process is shown on 12.4.



DC transistor acts in the same fashion as two connected diodes (11.4a). If both diodes are functional, transistor is functional as well as shown on 12.5. As you can see, probe A is connected to the base, and then probe B is connected first to the emitter, and then to the collector. In both cases, if the transistor is ok, “music” would have been heard. We then switch probe connections, A goes where B was connected to and vice versa, if there is no music now, everything is in order. So, transistor is faulty if speaker remains silent in the first two measurements, or if it “plays” in one of the second two measurements.

FET testing is done in similar fashion as testing the bipolar transistors, which is shown on 12.6.





One principle that is applicable when testing the photo resistors, photo transistors and diodes is NL-NM (or, No Light – No Music). Probe A is connected to the collector of the transistor, or diode's anode or one side of the photo resistor, and the other one is connected to transistor's emitter or diode's cathode or the other resistor's side and some kind of sound should be heard from the speaker. If this continues when the component is shadowed using your palm, everything is in functional order. We displayed graphically the method of testing photo sensitive components on 12.7.



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